Mandatory Assignment 1 - Dice Game 12

LEARNING OBJECTIVES

- Using decision statements
- Using repetition statements
- Using random numbers
- Using input validations

The most typical mistakes that students do:

- they do not validate input as per the requirements.
- they do not read the coding rules
- they do the assignment without reading about the theory
- they have not seen the assignment overview videos

Prerequisites

You should have completed all the Module 3 parts.

Task

You will create a program that simulates a variation of a popular game called "Shut the box". We will call this game "the dice game 12". **Please note, do NOT google "shut the box" game rules and try to implement that, instead follow the rules below. We mentioned "Shut the box" as an inspiration to this assignment."**

Begin by describing your structure in text form (pseudocode) in the comment header of the program so that an outsider can understand what the program does and how your logic/algorithm works.

The game consists of three dice rolls and involves getting 12 across three rolls of dice in each round. Each dice can only be rolled once per round. Example: If a player selects to roll dice #2, then the player cannot roll dice #2 again in the current round.

In each round, the player must be able to choose (by pressing on keyboard) between:

- 1 to roll the dice 1
- 2 to roll the dice 2
- 3 to roll the dice 3
- q to exit the game

Note the following:

- The program must randomly find a value on the selected dice and then calculate the score.
- The program should also present the number of wins and the number of rounds lost.
- The program should continue until the user chooses to cancel the game by pressing q on the keyboard.
- The definition of win is when the sum of the dice on all three rolls is 12, and
- The definition of loss is a sum exceeding 12 after all three dice have been rolled.
- If the sum after three rolls is less than 12, there will be no win or loss, but you go straight to the next round.
- Note dice values must be 1, 2, 3, 4, 5, or 6
- All three dices must be rolled before computing a result. In other words, result is checked only after each round and each round is equal to all three dices rolled.
  - For example: If first two dice results in 6 and 6 and a sum of 12, then it is should NOT be considered a loss (because any value on third roll is a loss) or it should NOT be considered a win because sum is already 12.

Example run of the program

```
Welcome to dice game 12. You must roll 1-3 dice and try to get the sum of 12 ...

Enter which dice you want to roll [1,2,3] (exit with q): 4
Sorry, that is an invalid entry. Try again. Valid entries are 1, 2, 3, and q

Enter which dice you want to roll [1,2,3] (exit with q): -1
Sorry, that is an invalid entry. Try again. Valid entries are 1, 2, 3, and q

Enter which dice you want to roll [1,2,3] (exit with q): w
Sorry, that is an invalid entry. Try again. Valid entries are 1, 2, 3, and q

Enter which dice you want to roll [1,2,3] (exit with q): string
Sorry, that is an invalid entry. Try again. Valid entries are 1, 2, 3, and q

Enter which dice you want to roll [1,2,3] (exit with q): 1
6 0 0 sum: 6 #win: 0 #loss: 0
Enter which dice you want to roll [1,2,3] (exit with q): 2
6 1 0 sum: 7 #win: 0 #loss: 0
Enter which dice you want to roll [1,2,3 ] (exit with q): 3
6 1 2 sum: 9 #win: 0 #loss: 0
You neither won nor lost the game.

Next round!
```

```
Enter which dice you want to roll [1,2,3] (exit with q): 1
6 0 0 sum: 6 #win: 0 #loss: 0
Enter which dice you want to roll [1,2,3] (exit with q): 1
Sorry, you have already rolled that dice. Try again
Enter which dice you want to roll [1,2,3] (exit with q): 2
6 3 0 sum: 9 #win: 0 #loss: 0
Enter which dice you want to roll [1,2,3] (exit with q): 3
6 3 3 sum: 12 #win: 1 #loss: 0
You won!!

Next round!

Enter which dice you want to roll [1,2,3] (exit with q): 2
0 6 0 sum: 6 #win: 0 #loss: 0
Enter which dice you want to roll [1,2,3] (exit with q): 1
3 6 0 sum: 9 #win: 0 #loss: 0
Enter which dice you want to roll [1,2,3] (exit with q): 3
3 6 5 sum: 14 #win: 1 #loss: 1
You lost!!

Next round!

Enter the dice you want to roll [1,2,3] (exit with q): q
#win: 1 #loss: 1
Game Over!
```

In this task, you must check/manage the user's entries and provide relevant feedback if an incorrect entry occurs. The program should not crash, no matter what the user enters. In each round, you can only roll each die once. Use constants for values that do not change.

How will you be graded?

- You will be graded and checked for functional and non-functional requirements
- Functional requirements
  - Code output is according to the specification given in the task description
  - Validation is important
    - Same dice number cannot be selected more than once in the same round
    - If the sum is 12, then the win count increments by 1
    - if the sum is less than 12, there is no change to the win and loss count
    - if the sum is more than 12, then the loss count increments by 1
    - Appropriate messages are shown
      - Error messages are shown, for example, from the sample above:
        - Sorry, you have already rolled that dice. Try again
        - Sorry, that is an invalid entry. Try again. Valid entries are 1, 2, 3, and q
      - Status and other helpful messages are shown, for example, from the sample above
        - Welcome to dice game 12. You must roll 1-3 dice and try to get the sum of 12 ...
        - Enter which dice you want to roll [1,2,3] (exit with q):
        - 3 6 5 sum: 14 #win: 1 #loss: 1
        - You won!!

        - You lost!!
        - You neither won nor lost the game.
        - Next round!
        - Game Over!
- Non Functional requirements
    - Strictly follow the coding rules
    - Use only the techniques that you have learnt so far in the course. Codegrade will report an error if you are using something you are not supposed to.

Stuck? Need a Guide?

You should try this first by yourself. If you are stuck or need more help. Then use the **Guide for Mandatory Assignment 1 - Dice Game 12 (https://canvas.ltu.se/courses/24507/pages/guide-for-mandatory-assignment-1-dice-game-12)**

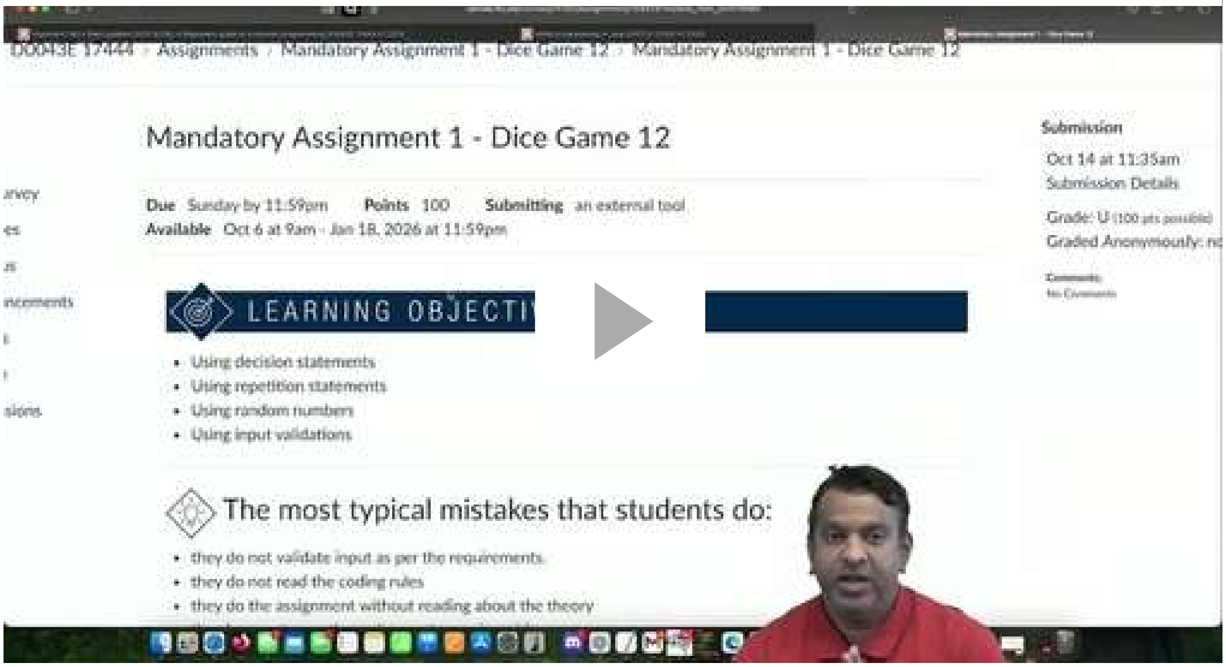
Getting help

Seems like many have not been reading my announcements. I have specifically requested how to seek help and what information to send me when you email or post on Discussions. I have also mentioned that it is not allowed to post answers to the Mandatory Assignments. If you need help, please see **How to get help the right way (https://canvas.ltu.se/courses/24507/pages/how-to-get-help-the-right-way)**

Videos

How to test your code

D0043E Testing your MA1 Code



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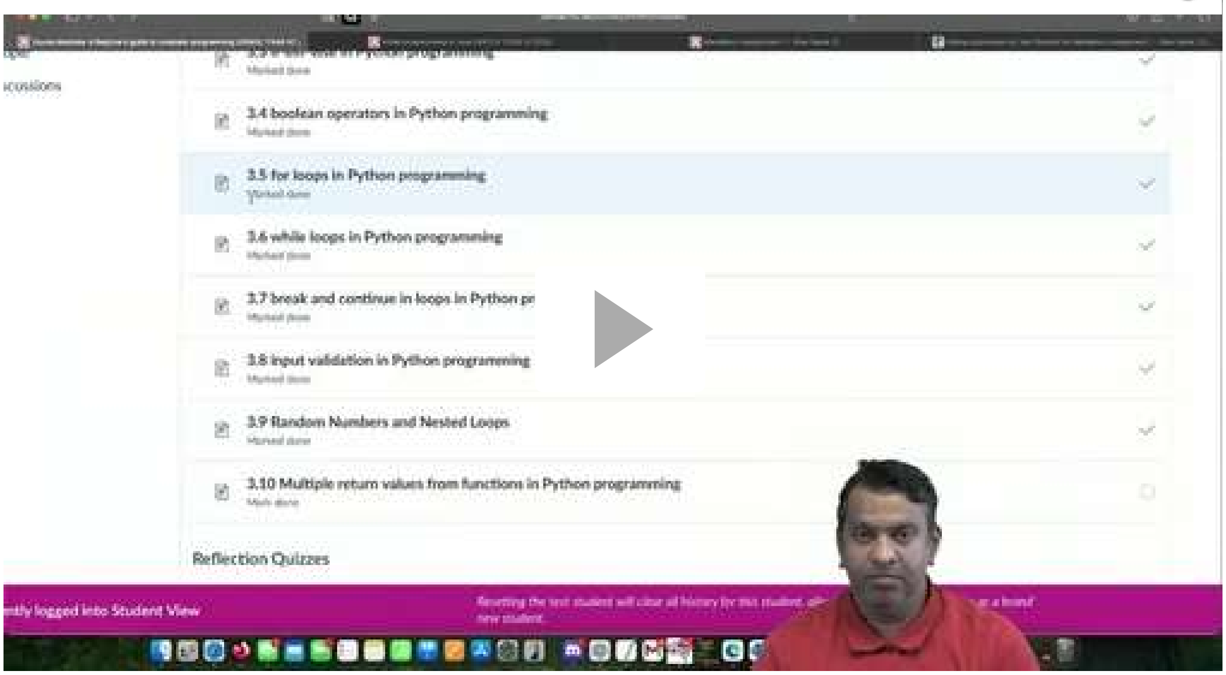
Hints and tips - step wise guide to solve the assignment

D0043E MA1 Overview



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